

Exercise: Evaluating a Probabilistic Forecaster

Predictions: Theory and Practice

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Part 1: Calibration for event probabilities

A forecasting platform records $n = 2000$ probabilistic predictions $\hat{p}$ for binary events $y \in \{0, 1\}$. Each question is labelled as *political* ($x = 1$) or non-political ($x = 0$). The data frame `dat` has the columns `p_hat`, `y`, and `political`.

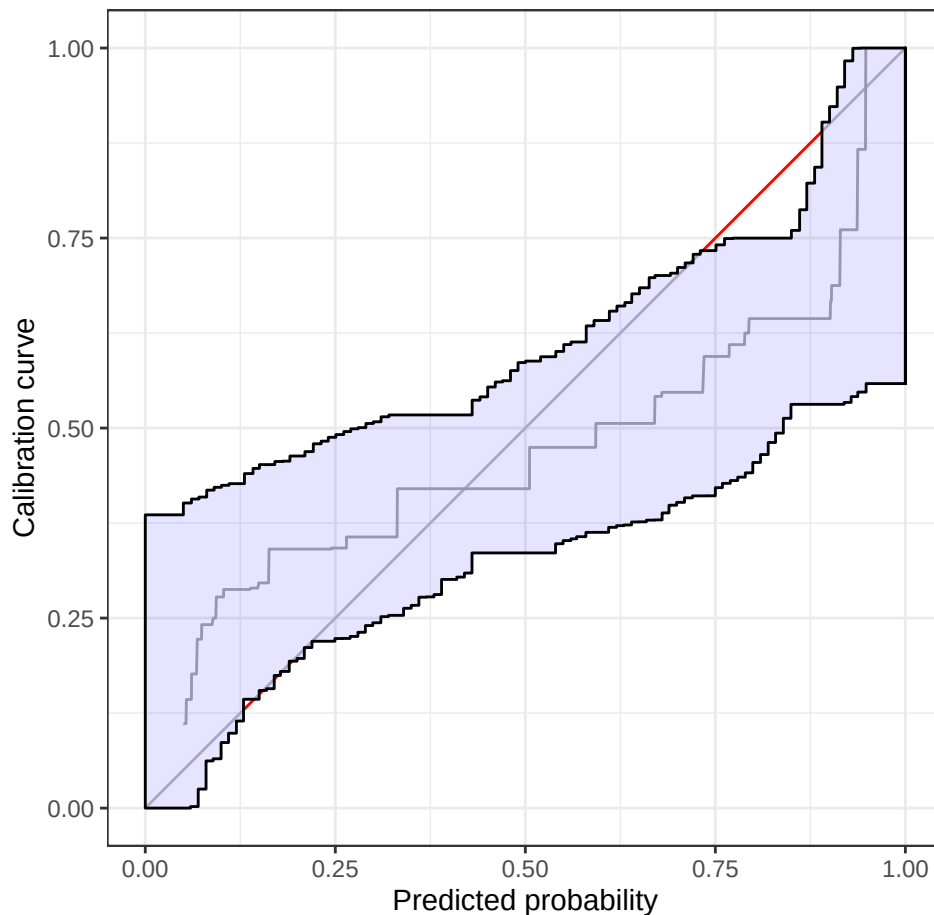
Your task is to evaluate the forecaster on this sample.

Recall from the lecture that a prediction $\hat{p}$ is calibrated for y if

$$E[y \mid \hat{p}] = \hat{p}.$$

We estimate the calibration curve $\hat{E}[y \mid \hat{p}]$ together with a 90% confidence band using the `calibrationband` package.

```
plot(calibration_bands(x = dat$p_hat, y = dat$y,  
                      alpha = 0.10, method = "round", digits = 2))
```



The diagonal is the reference line of perfect calibration. The shaded area is the 90% confidence band for $E[y \mid \hat{p}]$.

(a) Describe the deviation from calibration.

(b) Describe the evidence against calibration for $\hat{p} = 0.50$.

(c) Recalibrate a new prediction

The forecaster issues a new prediction of $\hat{p} = 0.75$. How would you recalibrate it? Read the answer off the plot, including a rough uncertainty range from the 90% band.

(d) Subgroup bias: political vs. non-political

To check whether the bias differs across question types, we run a linear regression of the prediction error $y - \hat{p}$ on the political indicator x :

$$y - \hat{p} = \beta_0 + \beta_1 \cdot x + \varepsilon, \quad x = \mathbb{1}\{\text{political question}\}.$$

```
fit <- lm(I(y - p_hat) ~ political, data = dat)
round(summary(fit)$coefficients,digits = 3)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.005	0.013	-0.354	0.723
political	-0.104	0.024	-4.269	0.000

Interpret the estimates of β_0 and β_1 . Be precise about what each coefficient measures and how it relates to (mis-)calibration.

Part 2: Calibration for real-valued targets

A second forecaster issues **probabilistic predictions for a real-valued target** y (e.g. next quarter's GDP growth, in percent). For each observation i the forecaster reports a predictive distribution $\hat{P}_i$. We have $n_2 = 500$ such forecasts in the data frame.

(a) Compute a single PIT value

For the first observation, the prediction was a uniform distribution with support $[0, 4]$,

$$\hat{P}_1 = U[0, 4],$$

and the observed value was $y_1 = 2$.

Compute the PIT value $PIT(y_1, \hat{P}_1) = F_{\hat{P}_1}(y_1)$. What does it mean?

(b) Compute a single PIT value

For the second observation, the prediction was a uniform distribution with support $[0, 3]$ ($\hat{P}_2 = U[0, 3]$) and the observed value was $y_2 = 0$.

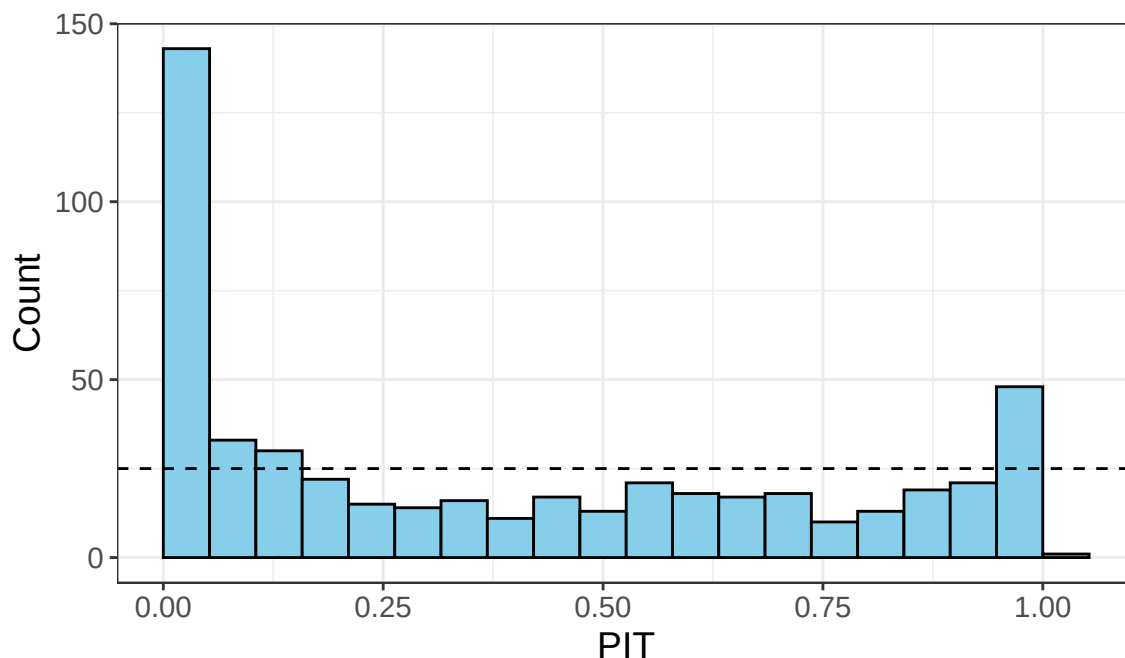
Compute the PIT value. What does it mean?

(c) Interpret the PIT histogram

Recall from the lecture that if the prediction is calibrated, the Probability Integral Transform

$$PIT_i = F_{\hat{P}_i}(y_i)$$

is uniformly distributed on $[0, 1]$. We compute the PIT values and plot their histogram (the dashed line shows the uniform reference level):



Describe the pattern of miscalibration visible in the histogram. What does it tell you about *where* the forecaster's predictive distribution is wrong — in its **location** (mean), in its **spread** (variance), or both? In which direction?

Part 3: Comparing two forecasters

A forecasting platform records $n = 2000$ probabilistic predictions from **two forecasters, A and B**, for binary events $y \in \{0, 1\}$. Each question is again labelled *political* (`political = 1`) or non-political (`political = 0`).

(a) Realized quadratic scores

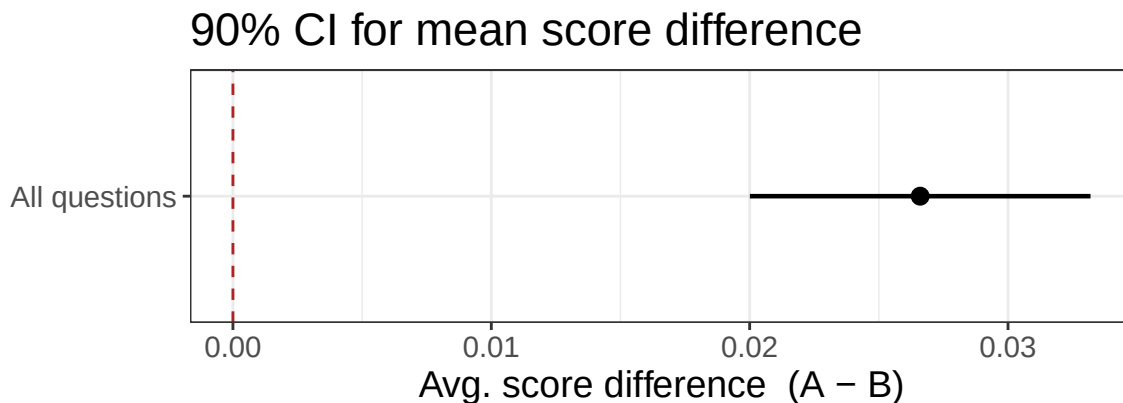
The **realized quadratic score** for a single prediction is

$$s_i(\hat{p}) = 1 - (y_i - \hat{p}_i)^2.$$

For the single observation $y = 1$, $\hat{p}^A = 0.1$, $\hat{p}^B = 0.5$, compute the score for each forecaster by hand.

(b) Score-difference plot with confidence interval — full sample

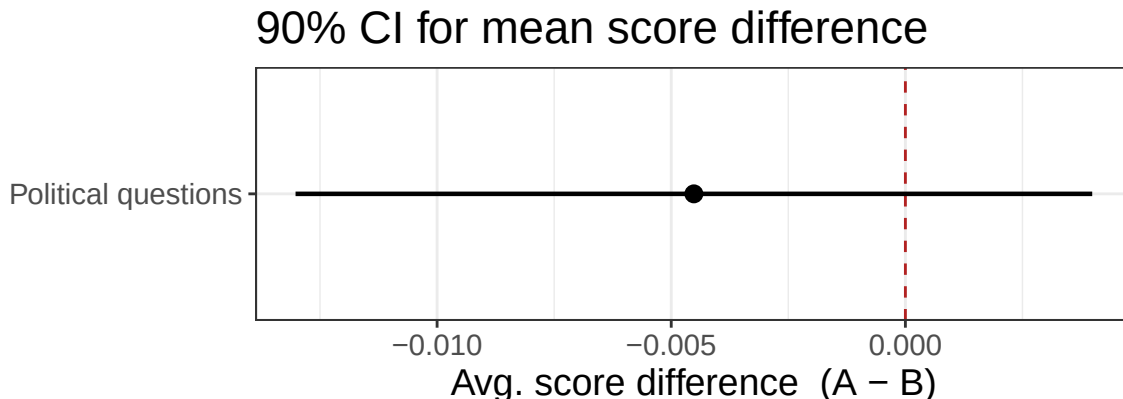
The plot below shows the **average score difference** $\bar{\Delta} = \frac{1}{n} \sum_i (s_i^A - s_i^B)$ with $s_i = 1 - (y_i - \hat{p}_i)^2$, together with a **90% confidence interval**.



Based on the plot, which forecaster is more accurate? Is the result statistically significant?

(c) Score-difference plot — political questions only

The same analysis restricted to **political questions** only:



Who appears more accurate on political questions? What does the confidence interval tell you about the strength of the evidence?